

# CBSE Class 10 Mathematics - Test Paper

Time Allowed: 3 Hours

Maximum Marks: 80

---

## General Instructions:

1. This question paper contains questions. All questions are compulsory.
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

## Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. The decimal expansion of  $\frac{147}{1200}$  will terminate after how many places of decimal?

- (A) (A) 2  
(B) (B) 3  
(C) (C) 4  
(D) (D) 5

*(simple)*

## Section B: Very Short Answer (2 marks each)

1. If  $A = 2^3 \times 5^2$  and  $B = 2^2 \times 5^3 \times 7$ , find the HCF of  $A$  and  $B$ .

*(simple)*

## Section C: Short Answer I (3 marks each)

1. Divide the polynomial  $p(x) = x^3 - 3x^2 + 5x - 3$  by  $g(x) = x^2 - 2$  using the division algorithm. Find the quotient  $q(x)$  and remainder  $r(x)$ .

*(medium)*

## Section D: Long Answer (4-5 marks each)

1. Solve the following pair of equations by reducing them to a pair of linear equations:  
 $\frac{2}{x} + \frac{3}{y} = 11$  and  $\frac{5}{x} - \frac{4}{y} = -7$  *(medium)*
2. A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. Find the usual speed of the train. (Note: Form a quadratic equation and solve it.) *(hard)*
3. In  $\triangle ABC$ ,  $P$  and  $Q$  are points on sides  $AB$  and  $AC$  respectively such that  $PQ \parallel BC$ . The area of  $\triangle APQ$  is  $16 \text{ cm}^2$  and the area of trapezium  $PBCQ$  is  $20 \text{ cm}^2$ . Find the ratio  $AP : PB$ . If  $AB = 12 \text{ cm}$ , find the lengths of  $AP$  and  $PB$ . *(hard)*